

To:
Whom it may concern

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Your Reference

Subject
Reference letter for Mr. Erwan Achat

I am a Senior Research Scientist in the AVIZ project team at Inria Saclay, France, and my research focuses on topics in scientific visualization, illustrative rendering, and interactive visualization techniques. In this function I am writing this reference letter for Erwan Achat, who I met both as a student in a class I taught and as an intern in our team.

I first got to know Mr. Achat in the academic year 2023–2024, when he took my “Introduction to Computer Graphics” class, that I teach every year at Polytech Paris-Saclay. In this class (tobias.isenberg.cc/graphics-intro/), the students learn the basics of the computer graphics rendering pipeline and, in a group project, implement an animated and interactive scene in OpenGL. In this implementation part, Mr. Achat’s group (two students) implemented a solar system animation (tobias.isenberg.cc/graphics-intro/uploads/Gallery/202324-team11.png), and were in the top third to top quarter of all projects in that class. Overall (i.e., including the exam), I would also say that Mr. Achat finished in the top third to top quarter of the class.

After having finished the class, Mr. Achat approached me for joining the AVIZ project team for the computer science research internship that is part of his curriculum at Polytech Paris-Saclay, and he worked with us from April 14 to August 29 in 2025. In this internship he specifically worked within our ANR-funded tVIST research project, in which we investigate data visualization scenarios and platforms that go beyond our traditional planar and rectangular screens.

Mr. Achat’s primary task in his internship was to develop a prototypical implementation of a small and potentially mobile non-planar visualization display for the use in an office context, with which the noise levels from different directions are visualized over time. For this purpose he carried out multiple research- and development-related tasks: he familiarized himself with the relevant academic literature on data visualization (literature search and study), investigated and proposed several possible solutions for a prototype, ordered relevant electronic components for the physical implementation of the project, implemented (both hardware and software) a directional sound level meter with analog microphones and a microcontroller, implemented the transmission of sound data to the visualization component in real time via WebSocket protocol, implemented several visualizations of the received data, (due to the lack of a viable hardware display solution that we could not acquire in time for his work) implemented a virtual 3D web application to render his visualization templates on a cylindric screen, with the ability to switch between randomly generated data and actual sensor data coming from the WebSocket connection he had realized. Together with another intern working on a related project, he also contributed to a machine learning solution for understanding how recoded sound can be mapped to noise levels. Beyond his actual realization, he also suggested better sound processing and visualization system (mathematical basis, hardware components, and algorithm)—yet this solution was not possible to realize due to the constraints of our organization’s own administrative processes (e.g., ordering the needed more advanced materials) and the short time frame of Mr. Achat’s internship.

Overall, with this project Mr. Achat realized a nice and creative design (as well as—within the mentioned constraints—a fully implemented solution) for our tVIST research project, which will now serve as the basis and inspiration for our future developments for the PhD students who started just this month (October 2025) in our research team. I also note that Mr. Achat communicated very well within the team and within our research group in English (our standard language of communication), and participated well in group activities.

Should you have further questions please do not hesitate to contact me.

Sincerely,

A handwritten signature in blue ink, appearing to read 'T. Isenberg', with a stylized flourish at the end.

Dr. Tobias Isenberg